Car Dekho Analysis
DIY Project 4

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Problem Statement of Car Dekho Analysis

Car Dekho's online used car market lists numerous used cars where sellers can create their listings with prices based on minimal knowledge of the market trends, making them either over-priced or under-priced leading to either customer mistrust in the car listings on the marketplace and further decreasing the pace of selling of vehicles. There is a necessity to perform an analysis of all the used car listings listed on the website of Car Dekho in the past which are expected to analyse what drives a particular car's selling price in the future-Age of vehicle, Brand of the vehicle, KM driven on the vehicle, Vehicle Fuel Type, Vehicle Gear box and Previous Owners

Project Description of the Car Dekho Analysis

This project performs a Data Analysis (EDA) on the Car Dekho used car dataset, which contains the following attributes:

Car_Name : Name/model of the car

Year : Year of purchase

Selling_Price : Price at which the car is being sold (target variable, in lakhs)

Present_Price : Current ex-showroom price (in lakhs)

Kms_Driven : Total kilometers driven

Fuel_Type: Petrol / Diesel / CNG

Seller_Type: Dealer / Individual

Transmission: Manual / Automatic

Owner :Number of previous owners

Project Objectives

1) From which manufacturing year to which manufacturing year vehicles are present in this data ?

2) What is the lowest price to which a vehicle is sold ?

3) What is the highest price to which a vehicle is sold ?

4) How Many records are there in this data ?

5) Are there any missing records in this data ?

6) How many different vehicles are present in this data ?

7) Which is the most sold vehicle in this data

8) Does the database include any CNG vehicle ? if yes how many of them are there ?

9) How many vehicles here are for sell from individuals directly

10) Does this database contain auto transmission vehicles ? If yes how many of them are there?

11) How many single person owned vehicles are there in the database

12) Which is the most and least cost depreciated vehicle in data ?

13) Which brands/vehicles are less affected by cost depreciation?

14) Are there any factors that affect cost depreciation?

15) Is the selling price affected by the Year and distance driven by the vehicle?

16) Can we get data about newer vehicles, i.e. vehicles manufactured after 2014?

17) Can we find data about only two-wheelers?

18) Which is the oldest bike sold here?

19) Which is the newest bike sold here?

20) Do you find any deal in two-wheelers that exceeded general expectations? What could be the reason?

21) Can we find data about only cars?

22) Which is the oldest car sold here?

23) Which is the newest car sold here?

24) What percentage of vehicles use Petrol, Diesel, and CNG fuel?

25) Which fuel type has the highest and lowest number of vehicles?

26) Do you find any deal in cars that exceeded general expectations? What could be the reason?

What is Data Wrangling ?

Data wrangling, a crucial phase in the data analysis process, involves preparing and cleaning raw data to make it suitable for analysis. This involves various tasks such as handling missing values, removing duplicates, and transforming data into a consistent format. Effective data wrangling ensures that the data is reliable and ready for exploration, enabling meaningful insights to be derived from it.

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)is a vital phase in data analysis involving the examination and visualization of data to uncover patterns ,trends, and insights.in the context of an Airbnb data analysis project, EDA allows researchers to gain a Comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary stastics, data visualization ,and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, uiding subsequent analytical decisions.

End Users of this Project

Car Dekho Platform (Business Team): To enable making data-backed price suggestions to the sellers on listing.

Individual Car Sellers: to be able to list their car at fair competitive, data-backed prices.

Car Buyers: to check if the asked prices in the market are reasonable or not when viewing the car.

Used Car Dealers/Partner Showrooms: to get their pricing benchmarked against market.

Marketing & Growth Teams (Car Dekho): To check the brand/model preference and decide to promote accordingly.

Data Science/Analytics Team (Car Dekho): To build and improve upon the models for pricing/recommendation of the cars.

Students/Analysts: As a personal portfolio project showcasing a real E-commerce/Marketplace scenario.

Technologies Used In this Project

Hardware Requirements

The hardware requirements for performing the Car Dekho Analysis are:

RAM – 8.00 GB

Operating System – Windows 11

Processor – Intel(R) Core(TM) i3-1115g4

Processor speed – 3.00 GHz

Software Requirements

The software requirements for performing the Car Dekho Analysis are:

Programming Language – Python

Development Environment / IDE – Jupyter Notebook

Data Analysis Libraries – Pandas, NumPy

Data Visualization Libraries – Matplotlib, Seaborn

Dataset Format – CSV

Result of the Project

1) From which manufacturing year to which manufacturing year vehicles are present in this data ?

[27]:
```
print("Oldest Year:",df["Year"].min())
print("Newest Year:",df["Year"].max())
```

```
Oldest Year: 2003
Newest Year: 2018
```

Vehicles manufactured from 2003 to 2018 are present in the dataset.

2) What is the lowest price to which a vehicle is sold ?

[26]:
```
lowest_price = df["Selling_Price"].min()
df[df["Selling_Price"] == lowest_price]
```

[26]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
200	Bajaj Pulsar 150	2006	0.1	0.75	92233	Petrol	Individual	Manual	0

The lowest selling Price is 0.10 lakh

3) What is the highest price to which a vehicle is sold ?

[23]:
```
highest_price = df["Selling_Price"].max()
df[df["Selling_Price"] == highest_price]
```

[23]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
86	land cruiser	2010	35.0	92.6	78000	Diesel	Dealer	Manual	0

The highest selling Price is 35 lakh

4) How Many records are there in this data ?

[20]:
```
print("Number of Rows",df.shape[0])
```

```
Number of Rows 301
```

There are 301 records in the dataset.

[12]:
```
df.dtypes
```

[12]:
```
Car_Name          object
Year               int64
Selling_Price    float64
Present_Price    float64
Kms_Driven         int64
Fuel_Type         object
Seller_Type       object
Transmission      object
Owner              int64
dtype: object
```

5) Are there any missing records in this data ?

[13]:
```
df.isnull().sum()
```

[13]:
```
Car_Name         0
Year             0
Selling_Price    0
Present_Price    0
Kms_Driven       0
Fuel_Type        0
Seller_Type      0
Transmission     0
Owner            0
dtype: int64
```

There are no missing values in the dataset.

6) How many different vehicles are present in this data ?

```
[30]: print(df["Car_Name"].unique())
      df["Car_Name"].value_counts()
```

```
['ritz' 'sx4' 'ciaz' 'wagon r' 'swift' 'vitara brezza' 's cross'
 'alto 800' 'ertiga' 'dzire' 'alto k10' 'ignis' '800' 'baleno' 'omni'
 'fortuner' 'innova' 'corolla altis' 'etios cross' 'etios g' 'etios liva'
 'corolla' 'etios gd' 'camry' 'land cruiser' 'Royal Enfield Thunder 500'
 'UM Renegade Mojave' 'KTM RC200' 'Bajaj Dominar 400'
 'Royal Enfield Classic 350' 'KTM RC390' 'Hyosung GT250R'
 'Royal Enfield Thunder 350' 'KTM 390 Duke ' 'Mahindra Mojo XT300'
 'Bajaj Pulsar RS200' 'Royal Enfield Bullet 350'
 'Royal Enfield Classic 500' 'Bajaj Avenger 220' 'Bajaj Avenger 150'
 'Honda CB Hornet 160R' 'Yamaha FZ S V 2.0' 'Yamaha FZ 16'
 'TVS Apache RTR 160' 'Bajaj Pulsar 150' 'Honda CBR 150' 'Hero Extreme'
 'Bajaj Avenger 220 dtsi' 'Bajaj Avenger 150 street' 'Yamaha FZ  v 2.0'
 'Bajaj Pulsar  NS 200' 'Bajaj Pulsar 220 F' 'TVS Apache RTR 180'
 'Hero Passion X pro' 'Bajaj Pulsar NS 200' 'Yamaha Fazer '
 'Honda Activa 4G' 'TVS Sport ' 'Honda Dream Yuga '
 'Bajaj Avenger Street 220' 'Hero Splender iSmart' 'Activa 3g'
 'Hero Passion Pro' 'Honda CB Trigger' 'Yamaha FZ S '
 'Bajaj Pulsar 135 LS' 'Activa 4g' 'Honda CB Unicorn'
 'Hero Honda CBZ extreme' 'Honda Karizma' 'Honda Activa 125' 'TVS Jupyter'
 'Hero Honda Passion Pro' 'Hero Splender Plus' 'Honda CB Shine'
 'Bajaj Discover 100' 'Suzuki Access 125' 'TVS Wego' 'Honda CB twister'
 'Hero Glamour' 'Hero Super Splendor' 'Bajaj Discover 125' 'Hero Hunk'
 'Hero  Ignitor Disc' 'Hero  CBZ Xtreme' 'Bajaj  ct 100' 'i20' 'grand i10'
 'i10' 'eon' 'xcent' 'elantra' 'creta' 'verna' 'city' 'brio' 'amaze'
 'jazz']
```

```
[30]: Car_Name
      city                 26
      corolla altis        16
      verna                14
      fortuner             11
      brio                 10
                           ..
      Honda Activa 125      1
      Hero Hunk             1
      Hero  Ignitor Disc    1
      Hero  CBZ Xtreme      1
      Bajaj  ct 100         1
      Name: count, Length: 98, dtype: int64
```

98 different vehicles are present.

7) Which is the most sold vehicle in this data

```
[154]: df["Car_Name"].value_counts().head(1)
```

```
[154]: Car_Name
       city    26
       Name: count, dtype: int64
```

Honda City is the most frequently occurring vehicle, with 26 records.

8) Does the database include any CNG vehicle ? if yes how many of them are there ?

[47]:
```
cng_vehicles = df[df["Fuel_Type"] =="CNG"]
cng_vehicles
cng_vehicles.shape[0]
```

[47]: 2

Yes, there are 2 CNG vehicles.

9) How many vehicles here are for sell from individuals directly ?

[49]:
```
individuals = df[df["Seller_Type"] == "Individual"]
individuals
individuals.shape[0]
```

[49]: 106

106 vehicles are being sold directly by individuals.

10) Does this database contain auto transmission vehicles ? If yes how many of them are there?

[51]:
```
automatic = df[df["Transmission"] == "Automatic"]
automatic
automatic.shape[0]
```

[51]: 40

Yes, there are 40 automatic transmission vehicles.

11) How many single person owned vehicles are there in the database ?

[54]:
```
single_owner = df[df["Owner"]==0]
single_owner.shape[0]
```

[54]:
```
290
```

Yes, there are 40 automatic transmission vehicles.

12) Which is the most and least cost depreciated vehicle in data ?

```
[58]: df ["Depreciation"] = df["Present_Price"] - df["Selling_Price"]
      df[["Car_Name", "Selling_Price","Present_Price","Depreciation"]]
```

[58]:

	Car Name	Selling Price	Present Price	Depreciation
0	ritz	3.35	5.59	2.24
1	sx4	4.75	9.54	4.79
2	ciaz	7.25	9.85	2.60
3	wagon r	2.85	4.15	1.30
4	swift	4.60	6.87	2.27
...	...	...	...	...
296	city	9.50	11.60	2.10
297	brio	4.00	5.90	1.90
298	city	3.35	11.00	7.65
299	city	11.50	12.50	1.00
300	brio	5.30	5.90	0.60

301 rows × 4 columns

```
[ ]: most_depreciated = df.iloc[
         df["Depreciation"].idxmax()
         ]
     most_depreciated
```

The Land Cruiser has the highest depreciation

```
[62]: least_depreciated = df.iloc [
          df["Depreciation"].idxmin()
          ]
      least_depreciated
```

```
[62]: Car_Name          Honda Activa 4G
      Year                        2017
      Selling_Price               0.48
      Present_Price               0.51
      Kms_Driven                  4300
      Fuel_Type                 Petrol
      Seller_Type          Individual
      Transmission          Automatic
      Owner                         0
      Depreciation               0.03
      Name: 155, dtype: object
```

The Honda Activa 4G has the lowest depreciation.

13) Which brands/vehicles are less affected by cost depreciation?

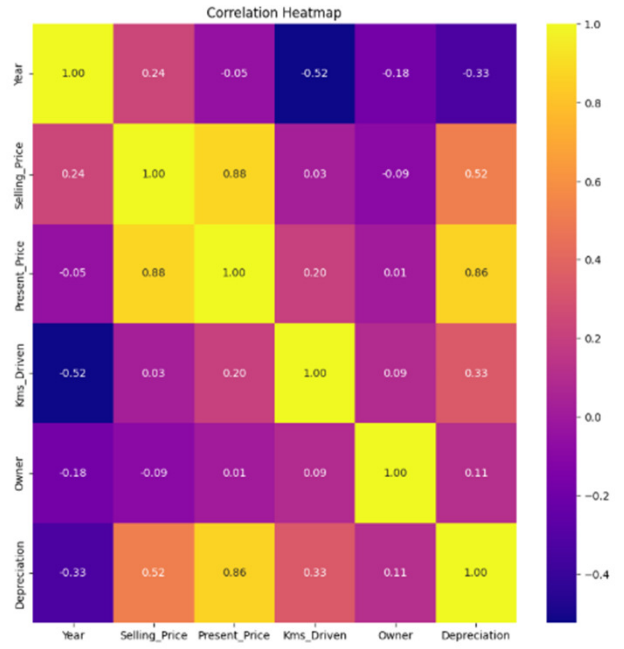
[82]:
```
Depreciation_Analysis = df.groupby("Car_Name")["Depreciation"].mean().sort_values()
Depreciation_Analysis.head(10)
```

[82]:
```
Car_Name
TVS Sport                   0.040
Honda Activa 4G             0.045
Hero Passion X pro          0.050
Bajaj Avenger 150           0.050
Honda Dream Yuga            0.060
Yamaha FZ 16                0.070
Activa 4g                   0.110
Hero Splender iSmart        0.115
Royal Enfield Bullet 350    0.120
UM Renegade Mojave          0.120
Name: Depreciation, dtype: float64
```

Vehicles such as TVS Sport,Honda Activa 4G ,Hero Passion X pro ,Bajaj Avenger 150, Honda Dream Yuga , Yamaha FZ 16,Activa 4g, Hero Splender iSmart,Royal Enfield Bullet 350, UM Renegade Mojave are less affected by Depreciation

14) Are there any factors that affect cost depreciation?

[91]:
```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10,10))
correlation = df.corr(numeric_only = True)
sns.heatmap(
    correlation,
    annot=True,
    cmap="plasma",
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```



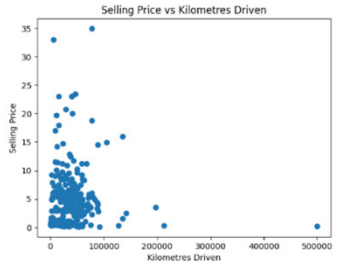
Year, kilometres driven, present price, ownership, transmission, fuel type, and market demand are important factors affecting depreciation.

15) Is the selling price affected by the Year and distance driven by the vehicle?

```
[109]: plt.scatter(df["Kms_Driven"], df["Selling_Price"])

       plt.xlabel("Kilometres Driven")
       plt.ylabel("Selling Price")
       plt.title("Selling Price vs Kilometres Driven")

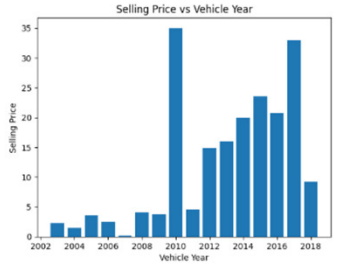
       plt.show()
```




```
[112]: plt.bar(df["Year"], df["Selling_Price"])

       plt.xlabel("Vehicle Year")
       plt.ylabel("Selling Price")
       plt.title("Selling Price vs Vehicle Year")

       plt.show()
```



Yes, the selling price is generally affected by both vehicle Year and kilometres driven.

16) Can we get data about newer vehicles, i.e. vehicles manufactured after 2014?

[113]:
```
newer_vehicles = df[df["Year"] > 2014]
newer_vehicles
```

[113]:

	Car Name	Year	Selling Price	Present Price	Kms Driven	Fuel Type	Seller Type	Transmission	Owner	Depreciation
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0	2.60
5	vitara brezza	2018	9.25	9.83	2071	Diesel	Dealer	Manual	0	0.58
6	ciaz	2015	6.75	8.12	18796	Petrol	Dealer	Manual	0	1.37
7	s cross	2015	6.50	8.61	33429	Diesel	Dealer	Manual	0	2.11
8	ciaz	2016	8.75	8.89	20273	Diesel	Dealer	Manual	0	0.14
...	...	...	...	...	...	...	...	...	...	...
295	city	2015	8.55	13.09	60076	Diesel	Dealer	Manual	0	4.54
296	city	2016	9.50	11.60	33988	Diesel	Dealer	Manual	0	2.10
297	brio	2015	4.00	5.90	60000	Petrol	Dealer	Manual	0	1.90
299	city	2017	11.50	12.50	9000	Diesel	Dealer	Manual	0	1.00
300	brio	2016	5.30	5.90	5464	Petrol	Dealer	Manual	0	0.60

147 rows × 10 columns

[117]:
```
newer_vehicles.shape[0]
```

[117]:
```
147
```

[119]:
```
newer_vehicles["Year"].value_counts().sort_index()
```

[119]:
```
Year
2015    61
2016    50
2017    35
2018     1
Name: count, dtype: int64
```

There are 147 vehicles manufactured after 2014.

17) Can we find data about only two-wheelers?

[137]:
```
bike_brands = [
    "Bajaj",
    "Hero",
    "Honda",
    "Royal Enfield",
    "Yamaha",
    "KTM",
    "TVS",
    "Suzuki",
    "Hyosung"
]
```

[138]:
```
two_wheelers = df[
    df["Car_Name"].str.startswith(
        tuple(bike_brands),
        na=False
    )
]

two_wheelers
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner	Depreciation
100	Royal Enfield Thunder 500	2016	1.75	1.90	3000	Petrol	Individual	Manual	0	0.15
102	KTM RC200	2017	1.65	1.78	4000	Petrol	Individual	Manual	0	0.13
103	Bajaj Dominar 400	2017	1.45	1.60	1200	Petrol	Individual	Manual	0	0.15
104	Royal Enfield Classic 350	2017	1.35	1.47	4100	Petrol	Individual	Manual	0	0.12
105	KTM RC390	2015	1.35	2.37	21700	Petrol	Individual	Manual	0	1.02
...	...	...	...	...	...	...	...	...	...	...
195	Bajaj ct 100	2015	0.18	0.32	35000	Petrol	Individual	Manual	0	0.14
197	Honda CB twister	2010	0.16	0.51	33000	Petrol	Individual	Manual	0	0.35
198	Bajaj Discover 125	2011	0.15	0.57	35000	Petrol	Individual	Manual	1	0.42
199	Honda CB Shine	2007	0.12	0.58	53000	Petrol	Individual	Manual	0	0.46
200	Bajaj Pulsar 150	2006	0.10	0.75	92233	Petrol	Individual	Manual	0	0.65

96 rows × 10 columns

[139]:
```
two_wheelers.shape[0]
```

[139]: 96

The dataset contains approximately 96 two-wheelers, including brands/models such as:

1. Bajaj
2. Hero
3. Honda motorcycles and scooters
4. Royal Enfield
5. Yamaha
6. KTM
7. TVS
8. Suzuki
9. Hyosung

18) Which is the oldest bike sold here?

[140]:
```
oldest_bike = two_wheelers.loc[
    two_wheelers["Year"].idxmin()
]

oldest_bike
```

[140]:
```
Car_Name         Hero Super Splendor
Year                            2005
Selling_Price                    0.2
Present_Price                   0.57
Kms_Driven                     55000
Fuel_Type                     Petrol
Seller_Type                Individual
Transmission                   Manual
Owner                               0
Depreciation                     0.37
Name: 189, dtype: object
```

The Hero Super Splendor (2005) is the oldest bike in the dataset.

19) Which is the newest bike sold here?

```
[148]: newest_car = cars.loc[
           cars["Year"].idxmax()
       ]

       newest_car
```

```
[148]: Car_Name         vitara brezza
       Year                      2018
       Selling_Price             9.25
       Present_Price             9.83
       Kms_Driven                2071
       Fuel_Type               Diesel
       Seller_Type             Dealer
       Transmission            Manual
       Owner                        0
       Depreciation              0.58
       Name: 5, dtype: object
```

The Vitara Brezza (2018) is the newest car in the dataset.

20) Do you find any deal in two-wheelers that exceeded general expectations? What could be the reason?

[142]:
```
two_wheelers = two_wheelers.copy()

two_wheelers["Depreciation"] = (
    two_wheelers["Present_Price"]
    - two_wheelers["Selling_Price"]
)

two_wheelers["Depreciation_Percentage"] = (
    two_wheelers["Depreciation"]
    / two_wheelers["Present_Price"]
) * 100
```

[143]:
```
two_wheelers.sort_values(
    "Depreciation_Percentage"
).head(10)
```

[143]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner	Depreciation	Depreciation_Percentage
126	Bajaj Avenger 220	2017	0.90	0.95	1300	Petrol	Individual	Manual	0	0.05	5.263158
155	Honda Activa 4G	2017	0.48	0.51	4300	Petrol	Individual	Automatic	0	0.03	5.882353
127	Bajaj Avenger 150	2016	0.75	0.80	7000	Petrol	Individual	Manual	0	0.05	6.250000
129	Yamaha FZ S V 2.0	2017	0.78	0.84	5000	Petrol	Individual	Manual	0	0.06	7.142857
102	KTM RC200	2017	1.65	1.78	4000	Petrol	Individual	Manual	0	0.13	7.303371
156	TVS Sport	2017	0.48	0.52	15000	Petrol	Individual	Manual	0	0.04	7.692308
100	Royal Enfield Thunder 500	2016	1.75	1.90	3000	Petrol	Individual	Manual	0	0.15	7.894737
128	Honda CB Hornet 160R	2017	0.80	0.87	3000	Petrol	Individual	Manual	0	0.07	8.045977
104	Royal Enfield Classic 350	2017	1.35	1.47	4100	Petrol	Individual	Manual	0	0.12	8.163265
131	Yamaha FZ 16	2015	0.75	0.82	18000	Petrol	Individual	Manual	0	0.07	8.536585

Some popular two-wheelers perform better than the general depreciation trend because of brand value and strong demand in the used-vehicle market.

21) Can we find data about only cars?

[145]:
```
cars = df[
    ~df["Car_Name"].str.startswith(
        tuple(bike_brands),
        na=False
    )
]

cars
```

[145]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner	Depreciation
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0	2.24
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0	4.79
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0	2.60
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0	1.30
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0	2.27
...	...	...	...	...	...	...	...	...	...	...
296	city	2016	9.50	11.60	33988	Diesel	Dealer	Manual	0	2.10
297	brio	2015	4.00	5.90	60000	Petrol	Dealer	Manual	0	1.90
298	city	2009	3.35	11.00	87934	Petrol	Dealer	Manual	0	7.65
299	city	2017	11.50	12.50	9000	Diesel	Dealer	Manual	0	1.00
300	brio	2016	5.30	5.90	5464	Petrol	Dealer	Manual	0	0.60

205 rows × 10 columns

[146]:
```
cars.shape[0]
```

[146]:
```
205
```

Approximately 205 records belong to cars

22) Which is the oldest car sold here?

[147]:
```
oldest_car = cars.loc[
    cars["Year"].idxmin()
]

oldest_car
```

[147]:
```
Car_Name              800
Year                 2003
Selling_Price        0.35
Present_Price        2.28
Kms_Driven         127000
Fuel_Type          Petrol
Seller_Type    Individual
Transmission       Manual
Owner                   0
Depreciation         1.93
Name: 37, dtype: object
```

The Maruti 800 (2003) is the oldest car in the dataset.

23) Which is the newest car sold here?

[148]:
```
newest_car = cars.loc[
    cars["Year"].idxmax()
]

newest_car
```

[148]:
```
Car_Name         vitara brezza
Year                      2018
Selling_Price             9.25
Present_Price             9.83
Kms_Driven                2071
Fuel_Type               Diesel
Seller_Type             Dealer
Transmission            Manual
Owner                        0
Depreciation              0.58
Name: 5, dtype: object
```

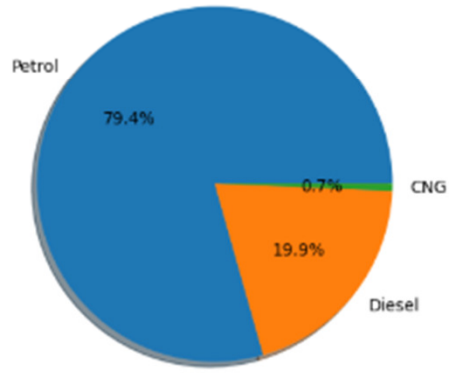
The Vitara Brezza (2018) is the newest car in the dataset.

24) What percentage of vehicles use Petrol, Diesel, and CNG fuel?

```
[129]: print(df.Fuel_Type.value_counts())

Fuel_Type
Petrol    239
Diesel     60
CNG         2
Name: count, dtype: int64

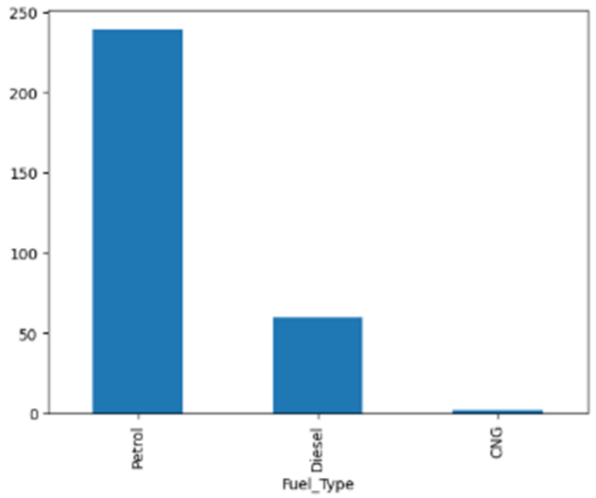
[132]: plt.pie(df.Fuel_Type.value_counts(),labels=["Petrol","Diesel","CNG"],shadow = True,autopct="%2.1f%%")
       plt.show()
```



Petrol has more consumption in the Pie Chart

25) Which fuel type has the highest and lowest number of vehicles?

```
[152]: (df.Fuel_Type.value_counts()).plot.bar()
       plt.show()
```



Petrol has more consumption in the Bar Chart

26) Do you find any deal in cars that exceeded general expectations? What could be the reason?

[149]:
```
cars = cars.copy()

cars["Depreciation"] = (
    cars["Present_Price"]
    - cars["Selling_Price"]
)

cars["Depreciation_Percentage"] = (
    cars["Depreciation"]
    / cars["Present_Price"]
) * 100
```

[150]:
```
cars.sort_values(
    "Depreciation_Percentage"
).head(10)
```

[150]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner	Depreciation	Depreciation_Percentage
80	corolla altis	2016	14.73	14.89	23000	Diesel	Dealer	Manual	0	0.16	1.074547
8	ciaz	2016	8.75	8.89	20273	Diesel	Dealer	Manual	0	0.14	1.574803
231	verna	2017	9.25	9.40	15001	Petrol	Dealer	Manual	0	0.15	1.595745
208	i20	2017	7.90	8.10	3435	Petrol	Dealer	Manual	0	0.20	2.469136
235	verna	2017	9.10	9.40	15141	Petrol	Dealer	Manual	0	0.30	3.191489
234	grand i10	2015	5.50	5.70	4492	Petrol	Dealer	Manual	0	0.20	3.508772
250	creta	2016	12.90	13.60	35934	Diesel	Dealer	Manual	0	0.70	5.147059
5	vitara brezza	2018	9.25	9.83	2071	Diesel	Dealer	Manual	0	0.58	5.900305
101	UM Renegade Mojave	2017	1.70	1.82	1400	Petrol	Individual	Manual	0	0.12	6.593407
83	innova	2015	12.50	13.46	38000	Diesel	Dealer	Manual	0	0.96	7.132244

[151]:

```
summary = pd.DataFrame({
    "Analysis": [
        "Oldest Year",
        "Newest Year",
        "Lowest Selling Price",
        "Highest Selling Price",
        "Total Records",
        "Unique Vehicles",
        "CNG Vehicles",
        "Individual Sellers",
        "Automatic Vehicles",
        "First Owner Vehicles",
        "Vehicles After 2014",
        "Two Wheelers",
        "Cars"
    ],

    "Result": [
        df["Year"].min(),
        df["Year"].max(),
        df["Selling_Price"].min(),
        df["Selling_Price"].max(),
        df.shape[0],
        df["Car_Name"].nunique(),
        (df["Fuel_Type"] == "CNG").sum(),
        (df["Seller_Type"] == "Individual").sum(),
        (df["Transmission"] == "Automatic").sum(),
        (df["Owner"] == 0).sum(),
        (df["Year"] > 2014).sum(),
        two_wheelers.shape[0],
        cars.shape[0]
    ]
})

print(summary)
```

```
                Analysis  Result
0            Oldest Year  2003.0
1            Newest Year  2018.0
2   Lowest Selling Price     0.1
3  Highest Selling Price    35.0
4          Total Records   301.0
5        Unique Vehicles    98.0
6           CNG Vehicles     2.0
7     Individual Sellers   106.0
8     Automatic Vehicles    40.0
9   First Owner Vehicles   290.0
10   Vehicles After 2014   147.0
11          Two Wheelers    96.0
12                  Cars   205.0
```

Future Scope

1. Predictive modelling : Turn the EDA in to a complete ML pipeline, like Linear regression, Random Forest, XG Boost or Gradient Boosting. Develop a reliable pricing predictor. The model should no longer be descriptive.
2. Live price suggestions: Build into a webapp / API hosted onto the car Dekho car listing website that will enable sellers to instantly receive a reasonable estimation of a car, even while they enter details.
3. Large & richer data: We should feed more information to the model, such as car age, car condition, accident history, maintenance history, insurance validity, varying demand on a city-to-city or region to region basis, and time of year or market dynamics etc.. For more efficient vehicle valuation.
4. In-depth analysis at Brand/ Model level: Carry out analysis at a more detailed level for various Brands/ Models. Identify models where value retains or depreciation is too much. Such an information is useful for the business to launch a market campaign or managing stock.
5. Market Trends and Time Series analysis: Study the variation of market prices at different times across few months / years; study variation in demand, cycles, effect of introduction of new vehicles, or a trend away from the prevalent fuel types. Eg. Growing preference for Electric cars.
6. Geographic/ Regional analysis: Include region/ location information. Market prices may differ across the country due to variation in the price on account of taxes, demand, and other factors like road condition.
7. Integration with market data: While using internal data it is advisable to include competitor pricing from other market places. This enables a better estimation of the vehicle.
8. Consumer behavior analysis: The pricing data could be used in conjunction with buyer searches and purchase information to help advise customers what sort of car the could be considering purchase, given the stated budget or preferences (recommendation engine).
9. Detect anomaly/ fraud in listing: While creating an estimation model there may be cases wherein an estimated price is considerably higher/ lower than the actual, which may represent data entry mistakes or fraud listing. These discrepancies could be identified from the modeling output.
10. Auto dashboard generation: Develop an auto refreshing business dashboard using Power BI / Tableau, so that Car Dekho's business users may take better decisions based on market changes and pricing data, without needing manual recasting or redevelopment.

Conclusion

This project was a study and interpretation of the Car Dekho used car dataset in order to identify and model those factors that affect the resale value of an automobile in the used car market. Through a comprehensive data cleaning process, analysis, and visualization, the project identified clear and understandable trends such as that an increase in the age of a car resulted in the depreciation of sale value, that increases in the distance travelled in the vehicle results in increased devaluation, and that differences in attributes such as fuel, transmission, seller type, and previous owners impact vehicle price. Most importantly, the analysis solidified that the original sale value of the vehicle and the age of the vehicle both represent the best and most powerful variables affecting resale value, while even additional influences such as previous owners and type of seller contribute significantly to price determination. Ultimately, the project showed how exploration of transaction data can replace speculation in evaluating the price of a used car and be the first step in providing value to consumers selling to or at Car Dekho and others in the used automobile market.

Thank You

GitHub Link

https://github.com/krishnamohanb21-svg/Car-Dekho-Analysis-DIY-Project-4-.git